3.1.1 The developing cell: cell division and cell differentiation

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the cell cycle	To include the processes taking place during interphase (G_1,S and G_2), mitosis and cytokinesis, leading to genetically identical daughter cells.
(b)	**(i)** the changes that take place in the nuclei and cells of animals and plants during mitosis	To include the changes in the nuclear envelope and the behaviour of the centrioles, spindle fibres, centromere, chromatids and chromosomes, and the formation of the cell plate in plant cells.
	(ii) the microscopic appearance of cells undergoing mitosis	To include the examination and drawing of stained sections or squashes of plant tissue and the identification of the stages observed. **PAG1** HSW4, HSW5, HSW6
(c)	the principal stages and features of apoptosis	To include cell shrinkage, nuclear condensation (pyknosis), blebs, nuclear fragmentation (karyorrhexis), the roles of phosphatidylserine and macrophages.
(d)	the importance of apoptosis and mitosis in growth and repair	To include examples of the roles of apoptosis in cell deletion and mitosis in cell addition.

(e)	**(i)**	the differentiation of stem cells into specialised cells	To include an appreciation of the differences between totipotent, pluripotent and multipotent stem cells, and the differentiation of bone marrow stem cells into specialised blood cells.
	(ii)	current applications and uses of stem cells.	To include the use of bone marrow stem cells.